STOCHASTIC PROCESSES · FALL 2026

Stationary Stochastic Processes

Strict-sense (SSS) and wide-sense (WSS) stationarity



Week 02 · Version 3.0

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This presentation has been formatted by Claude Opus5.5.

ROADMAP

Outline of Week 02

01 **Stochastic Processes X(t)**

02 **First-Order Statistics of X(t)**

03 **Higher-Order Statistics of X(t)**

04 **Mean, Autocorrelation & Autocovariance**

05 **Stationary Stochastic Processes**

06 **Strict-Sense Stationary (SSS) Processes**

07 **Wide-Sense Stationary (WSS) Processes**

Goal: know when the statistics of $X(t)$ do not depend on where we look in time, and what that buys us.

01

Stochastic Processes X(t)

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03 Higher-Order Statistics of X(t)

04 Mean, Autocorrelation & Autocovariance

05 Stationary Stochastic Processes

06 Strict-Sense Stationary (SSS) Processes

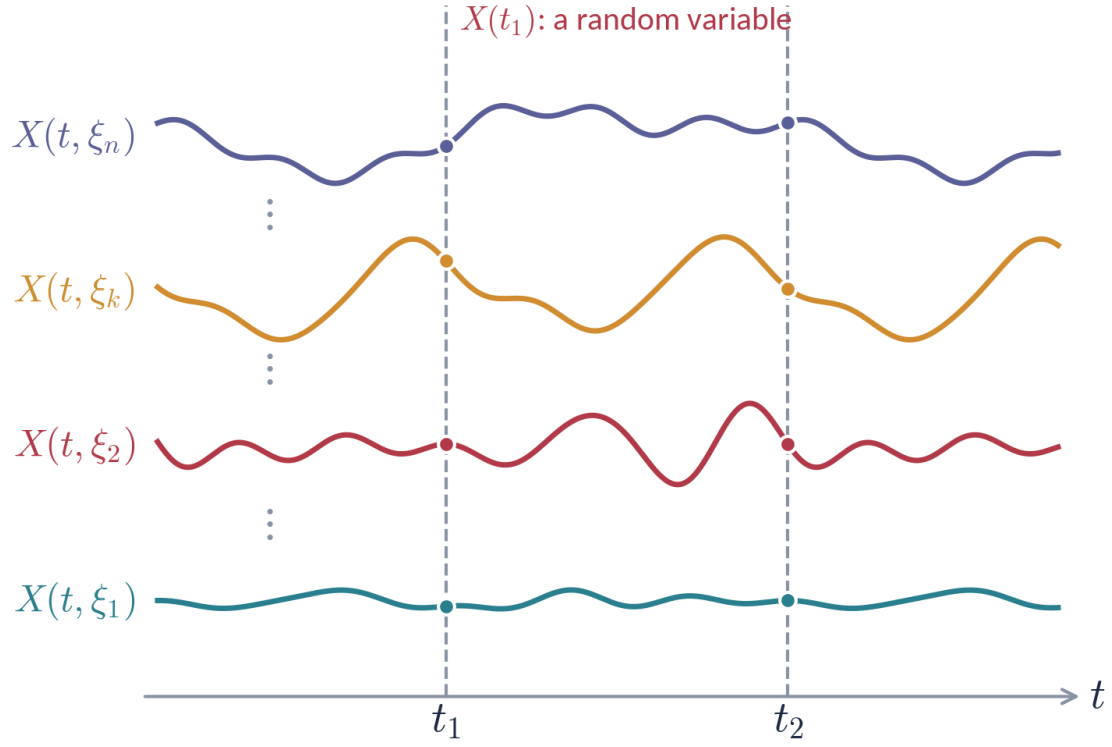
07 Wide-Sense Stationary (WSS) Processes

Stochastic Processes (recall)

- Let ξ denote the random outcome of an experiment, $\xi \in S$.
- To every outcome we assign a waveform $X(t,\xi)$ — a **sample path** (realization).
- The collection of all sample paths (the **ensemble**) forms the stochastic process $X(t)$.
- The index set of t and the values of X may each be continuous or discrete (countable or finite).

Notation

We write $X(t)$ for $X(t,\xi)$, dropping ξ when no confusion arises.

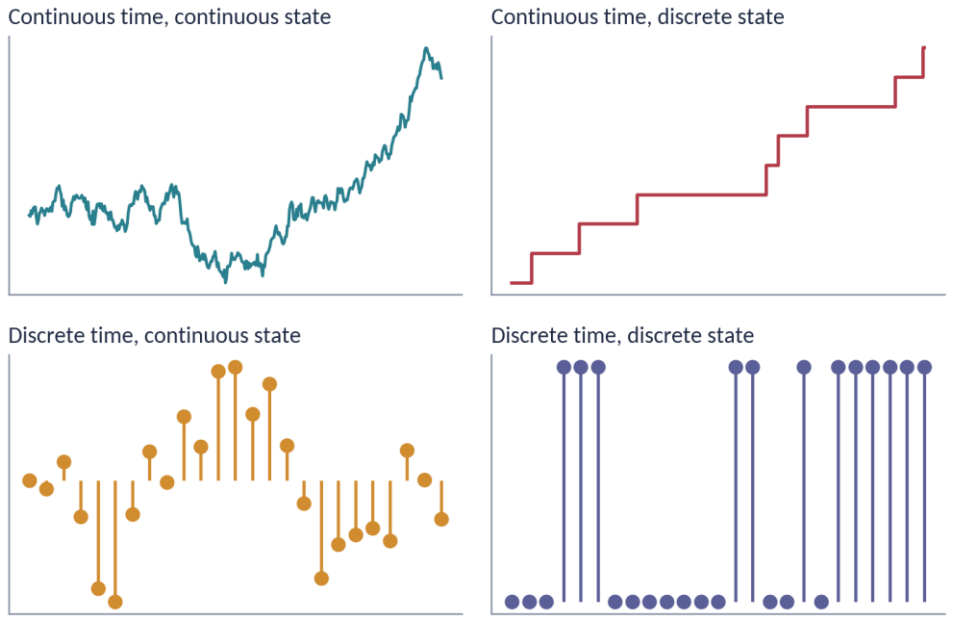


Two Views of the Same Object

	t fixed	t variable
ξ fixed	a number $X(t_1,\xi_i)$	a time function (sample path)
ξ variable	a random variable $X(t_1)$	the stochastic process $X(t)$

- For fixed ξ_i, $X(t,\xi_i)$ is a **specific time function**.
- For fixed t_1, $X_1=X(t_1,\xi)$ is a **random variable**.
- The ensemble of all realizations over time is the **stochastic process**.

Classification by time index and state space



e.g. thermal noise · Poisson counts · AR(1) sequence · Bernoulli sequence

02

First-Order Statistics of X(t)

01 Stochastic Processes X(t)

02 **First-Order Statistics of X(t)**

03 Higher-Order Statistics of X(t)

04 Mean, Autocorrelation & Autocovariance

05 Stationary Stochastic Processes

06 Strict-Sense Stationary (SSS) Processes

07 Wide-Sense Stationary (WSS) Processes

First-Order CDF and PDF

For a fixed t, $X(t)$ is a random variable. Its distribution function is

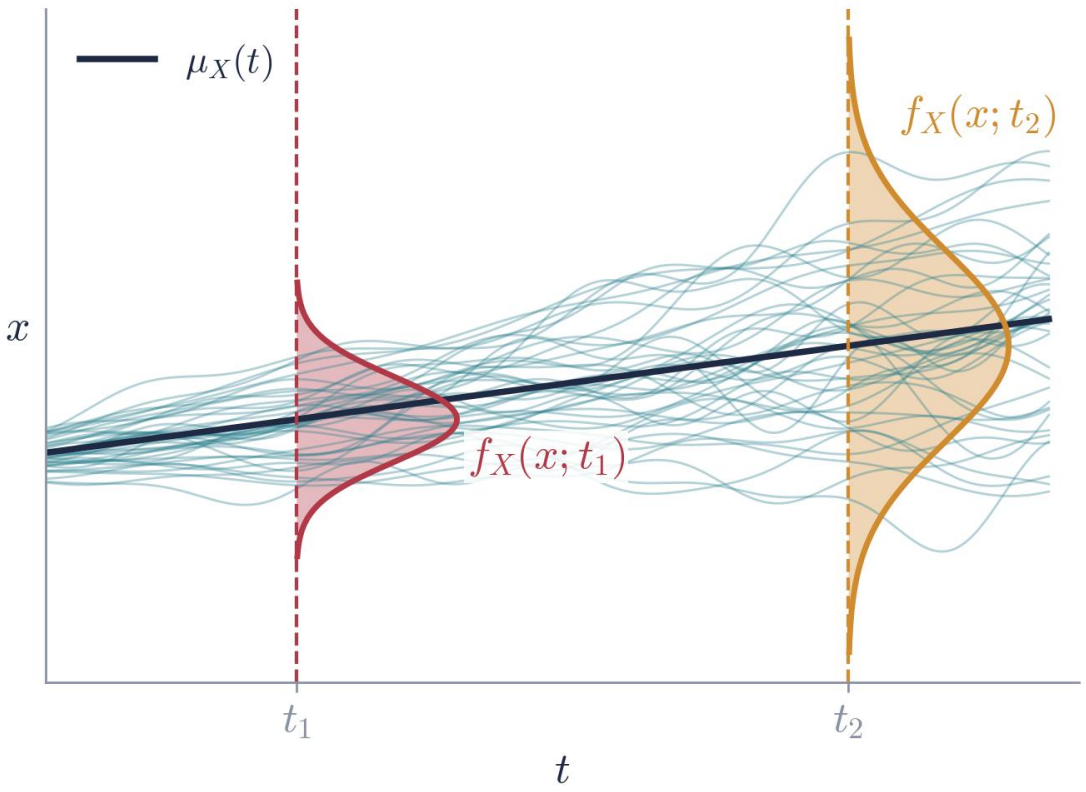
First-order distribution

$$F_X(x;t) = P\{X(t) \le x\}$$

and the **first-order pdf** of the process is

$$f_X(x;t) = \frac{\partial F_X(x;t)}{\partial x}$$

- $F_X(x;t)$ generally depends on t: a different t gives a different random variable.
- First-order statistics describe each time instant **in isolation** — they say nothing about how $X(t_1)$ and $X(t_2)$ are related.



The first-order pdf is a "slice" of the ensemble at one time; here both the mean and the spread change with t.

03

Higher-Order Statistics of X(t)

01 Stochastic Processes X(t)

02 First-Order Statistics of X(t)

03 **Higher-Order Statistics of X(t)**

04 Mean, Autocorrelation & Autocovariance

05 Stationary Stochastic Processes

06 Strict-Sense Stationary (SSS) Processes

07 Wide-Sense Stationary (WSS) Processes

Second-Order CDF and PDF

For $t{=}t_1$ and $t{=}t_2$, $X(t)$ gives two random variables $X_1{=}X(t_1)$ and $X_2{=}X(t_2)$. Their joint distribution is

Second-order distribution and density

$$F_X(x_1, x_2; t_1, t_2) = P\{X(t_1) \le x_1,\ X(t_2) \le x_2\}$$

$$f_X(x_1, x_2; t_1, t_2) = \frac{\partial^2 F_X(x_1, x_2; t_1, t_2)}{\partial x_1\, \partial x_2}$$

- The second-order density captures the **dependence** between the process at two instants.
- It must be consistent with the first-order density (marginalization):

$$f_X(x_1; t_1) = \int_{-\infty}^{\infty} f_X\,(x_1, x_2; t_1, t_2)\, dx_2$$

n-th Order Distributions

n-th order density

For any n time instants $t_1,\ldots,t_n$, the joint density of $X(t_1),\ldots,X(t_n)$

$$f_X(x_1, x_2, \ldots, x_n;\ t_1, t_2, \ldots, t_n)$$

is the **n-th order density** of the process $X(t)$.

- **Complete specification** of $X(t)$ requires $f_X(x_1,\ldots,x_n;t_1,\ldots,t_n)$ for **all** t_i, $i=1,\ldots,n$, and for **all** n — an almost impossible task in practice.
- These densities are not arbitrary: they must be **consistent** — symmetric under permuting the pairs (x_i,t_i), and each lower-order density must follow by integrating out variables (Kolmogorov consistency).
- In practice we therefore work with a few **moments**: the mean and the autocorrelation. This motivates the next section.

04

Mean, Autocorrelation & Autocovariance

01 Stochastic Processes X(t)

02 First-Order Statistics of X(t)

03 Higher-Order Statistics of X(t)

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05 Stationary Stochastic Processes

06 Strict-Sense Stationary (SSS) Processes

07 Wide-Sense Stationary (WSS) Processes

Mean and Autocorrelation

Mean (expected value) of a stochastic process

$$\mu_X(t) \triangleq E\{X(t)\} = \int_{-\infty}^{\infty} x \; f_X(x;t)\, dx$$

In general, the mean of a process can **depend on the time index** t.

Autocorrelation function

$$R_{XX}(t_1,t_2) \triangleq E\{X(t_1)X^*(t_2)\} = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} x_1\, x_2^*\, f_X(x_1,x_2;t_1,t_2)\, dx_1\, dx_2$$

It measures the **interrelationship** between the random variables X_1=$X(t_1)$ and X_2=$X(t_2)$ generated by the process. For real processes the conjugate * can be dropped.

Properties of the Autocorrelation Function

1. Hermitian symmetry

$$R_{XX}(t_1, t_2) = R^*_{XX}(t_2, t_1) = [E\{X(t_2)X^*(t_1)\}]^*$$

2. Average instantaneous power

$$R_{XX}(t, t) = E\{|X(t)|^2\} \geq 0$$

3. Non-negative definiteness — for any n, any $t_1,\ldots,t_n$ and any constants $\{a_i\}_{i=1}^n$

$$\sum_{i=1}^{n}\sum_{j=1}^{n} a_i\, a_j^*\, R_{XX}(t_i, t_j) \geq 0$$

Proof: let $Y=\sum_{i=1}^{n} a_i X(t_i)$. Then $E\{|Y|^2\}=\sum_i\sum_j a_i a_j^* E\{X(t_i)X^*(t_j)\}\geq 0$.

Autocovariance and Correlation Coefficient

Autocovariance

$$C_{XX}(t_1, t_2) = E\{[X(t_1) - \mu_X(t_1)][X(t_2) - \mu_X(t_2)]^*\} = R_{XX}(t_1, t_2) - \mu_X(t_1)\,\mu_X^*(t_2)$$

Its diagonal is the variance of the process: $\sigma_X^2(t)=C_{XX}(t,t)$.

Correlation coefficient

$$\rho_{XX}(t_1, t_2) = \frac{C_{XX}(t_1, t_2)}{\sqrt{C_{XX}(t_1, t_1)\,C_{XX}(t_2, t_2)}} = \frac{C_{XX}(t_1, t_2)}{\sigma_X(t_1)\,\sigma_X(t_2)}$$

- By the Cauchy–Schwarz inequality, $|\rho_{XX}(t_1,t_2)|\leq 1$.
- ρ_{XX} is a normalized, unit-free measure of how linearly related $X(t_1)$ and $X(t_2)$ are.

Example 1 — The Integrator

Let the random variable z be the integral of the process over $[-T,T]$:

$$z = \int_{-T}^{T} X(t)\, dt$$

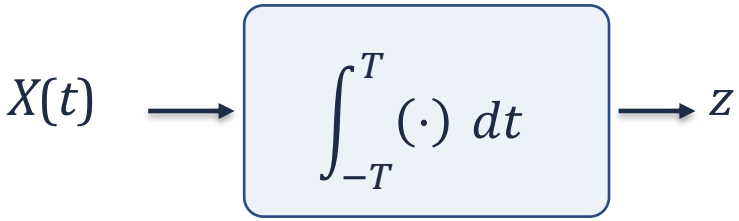
Mean (expectation and integration commute):

$$E\{z\} = \int_{-T}^{T} \mu_X(t)\, dt$$

Second moment:

$$E\{|z|^2\} = E\left\{\int_{-T}^{T}\int_{-T}^{T} X(t_1)X^*(t_2)\, dt_1\, dt_2\right\}$$

$$= \int_{-T}^{T}\int_{-T}^{T} R_{XX}(t_1,t_2)\, dt_1\, dt_2$$



Coming back later

For a WSS input this double integral collapses to a single integral over $\tau=t_1-t_2$ — see the integrator revisited in Section 07.

Example 2 — Sinusoid with Random Phase

$$X(t) = a\cos(\omega_0 t + \varphi), \qquad \varphi \sim U(0,2\pi)$$

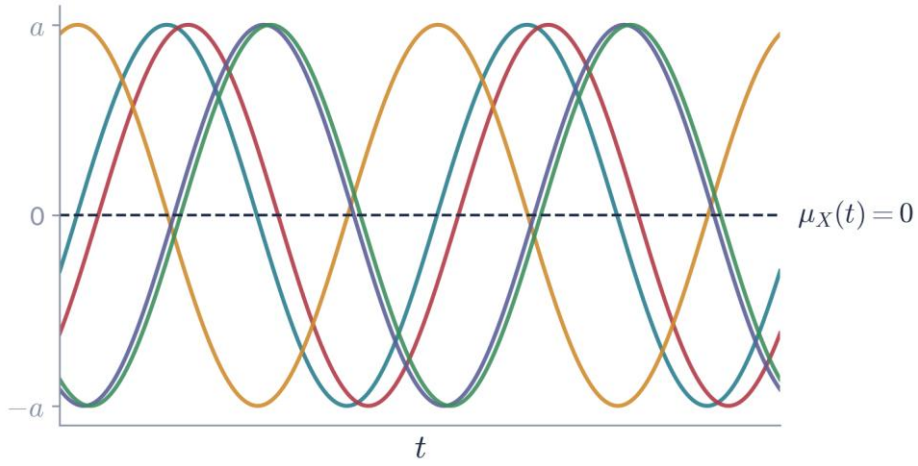
$$\mu_X(t) = E\{X(t)\} = \frac{a}{2\pi}\int_0^{2\pi} \cos(\omega_0 t + \varphi)\, d\varphi = 0$$

$$R_{XX}(t_1,t_2) = a^2 E\{\cos(\omega_0 t_1 + \varphi)\cos(\omega_0 t_2 + \varphi)\}$$

$$= \frac{a^2}{2} E\{\cos\omega_0(t_1 - t_2) + \cos(\omega_0(t_1 + t_2) + 2\varphi)\}$$

$$E\{\cdot\}=0$$

$$R_{XX}(t_1,t_2) = \frac{a^2}{2}\cos\omega_0(t_1 - t_2), \qquad \sigma_X^2 = R_{XX}(t,t) = \frac{a^2}{2}$$



Observe

The mean is constant and R_{XX} depends only on $t_1 - t_2$. We will call such a process **wide-sense stationary**.

Example 3 — Random Line

$$X(t) = At + b, \qquad A \sim N(0,1),\ b \text{ constant}$$

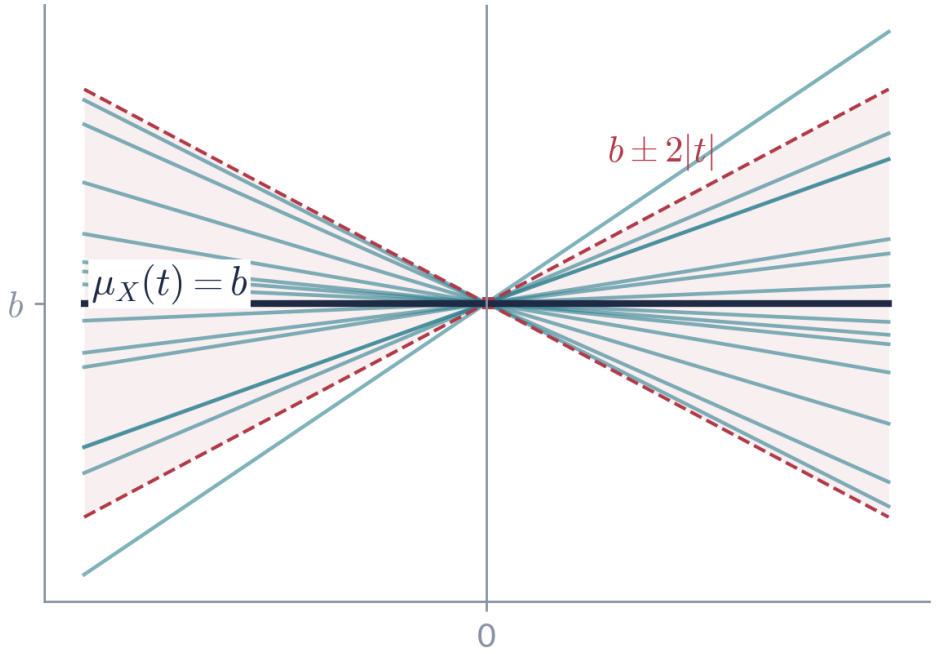
$$\mu_X(t) = E\{At + b\} = b$$

$$R_{XX}(t_1, t_2) = E\{(At_1 + b)(At_2 + b)\}$$

$$= E\{A^2\}t_1 t_2 + b\,E\{A\}(t_1 + t_2) + b^2 = t_1 t_2 + b^2$$

$$C_{XX}(t_1, t_2) = t_1 t_2, \qquad \sigma_X^2(t) = t^2$$

$$\rho_{XX}(t_1, t_2) = \frac{t_1 t_2}{|t_1|\,|t_2|} = \operatorname{sgn}(t_1 t_2) = \pm 1$$



Observe

Constant mean, but the variance t^2 grows with $|t|$ and R_{XX} is **not** a function of $t_1 - t_2$ alone: the process is not stationary.

05

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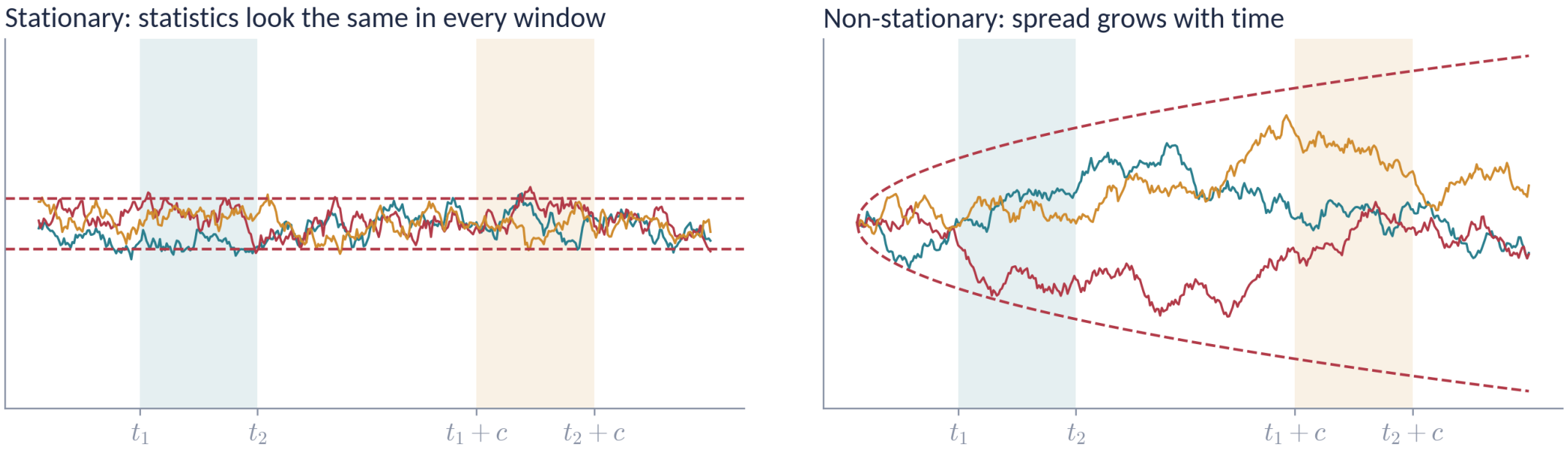
05 **Stationary Stochastic Processes**

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Stationary Stochastic Processes

- **Stationary processes** have statistical properties that are **invariant to a shift** of the time origin.
- **First-order** stationarity: $X(t_i)$ and $X(t_i{+}c)$ have the same statistics for any c.
- **Second-order** stationarity: the pairs $\{X(t_1),X(t_2)\}$ and $\{X(t_1{+}c),X(t_2{+}c)\}$ have the same joint statistics for any c.



Left: a stationary AR(1)-type process — the ±2σ band is flat. Right: a random walk — the band widens like $\sqrt{t}$.

06

Strict-Sense Stationary (SSS) Processes

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06 **Strict-Sense Stationary (SSS) Processes**

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Strict-Sense Stationarity: Definition

In strict terms, the statistical properties are governed by the joint pdf. A process is **n-th order strict-sense stationary** if, for any c,

n-th order SSS

$$f_X(x_1, x_2, \dots, x_n;\ t_1, t_2, \dots, t_n) \equiv f_X(x_1, x_2, \dots, x_n;\ t_1 + c,\ t_2 + c, \dots, t_n + c)$$

The left side is the joint density of $X_1=X(t_1),\dots,X_n=X(t_n)$; the right side is the joint density of $X_1'=X(t_1+c),\dots,X_n'=X(t_n+c)$.

Strict-sense stationary (SSS)

$X(t)$ is **SSS** if the equality above holds for all t_i, $i=1,\dots,n$, for all $n=1,2,\dots$ and for any c.

First-Order SSS

For a **first-order** strict-sense stationary process,

$$f_X(x;t) \equiv f_X(x;t+c) \quad \text{for any } c.$$

In particular, choosing $c=-t$ gives

$$f_X(x;t) = f_X(x;0) \triangleq f_X(x)$$

i.e. the first-order density of $X(t)$ is **independent of t**. Consequently

$$E\{X(t)\} = \int_{-\infty}^{\infty} x\ f_X(x)\,dx = \mu \quad \text{(a constant)}$$

$$\sigma_X^2(t) = \sigma^2 \quad \text{(a constant)}$$

Simple example

An i.i.d. sequence $X[n]$ is SSS of every order: its joint pdf $\prod_i f(x_i)$ does not involve the time indices at all.

Remark

Any function of a first-order moment, e.g. $E\{g(X(t))\}$, is then constant in time.

Second-Order SSS

Similarly, for a **second-order** strict-sense stationary process

$$f_X(x_1, x_2;\, t_1, t_2) \equiv f_X(x_1, x_2;\, t_1 + c,\, t_2 + c) \quad \text{for any } c.$$

Choosing $c = -t_2$:

$$f_X(x_1, x_2;\, t_1, t_2) \equiv f_X(x_1, x_2;\, t_1 - t_2)$$

The second-order density depends only on the **difference of the time indices** $\tau = t_1 - t_2$. Hence

Autocorrelation of a second-order SSS process

$$R_{XX}(t_1, t_2) = \int\int x_1 x_2^* f_X(x_1, x_2;\, \tau = t_1 - t_2)\, dx_1 dx_2 = R_{XX}(t_1 - t_2) \triangleq R_{XX}(\tau)$$

$$R_{XX}(\tau) = R_{XX}^*(-\tau)$$

From SSS to a Weaker Notion

- The results for μ and $R_{XX}(\tau)$ are **consequences** of first- and second-order strict-sense stationarity.
- n-th order SSS implies k-th order SSS for every $k \leq n$ (integrate out the extra variables).
- However, the basic conditions for n-th order stationarity — equality of **all** joint densities — are **usually difficult to verify**.
- We therefore often resort to a looser definition that only involves the first two moments: **Wide-Sense Stationarity (WSS)**.

SSS asks for …

- equal pdfs of every order
- under every time shift
- i.e. the whole probability law is shift-invariant

WSS asks only for …

- constant mean
- autocorrelation depending on τ only
- i.e. shift-invariance of moments up to order 2

07

Wide-Sense Stationary (WSS) Processes

01 Stochastic Processes X(t)

02 First-Order Statistics of X(t)

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04 Mean, Autocorrelation & Autocovariance

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06 Strict-Sense Stationary (SSS) Processes

07 **Wide-Sense Stationary (WSS) Processes**

Wide-Sense Stationarity: Definition

Wide-sense stationary (WSS)

A process $X(t)$ with $E\{|X(t)|^2\}<\infty$ is **wide-sense stationary** if

(i) $E\{X(t)\} = \mu$ (a constant)

(ii) $E\{X(t_1)X^*(t_2)\} = R_{XX}(t_1 - t_2)$ for all t_1, t_2

- For WSS processes the mean is constant and the autocorrelation depends only on the time difference. We write

$$R_{XX}(\tau) = E\{X(t+\tau)X^*(t)\}, \qquad C_{XX}(\tau) = R_{XX}(\tau) - |\mu|^2$$

- These conditions say **nothing about the pdfs**; they only constrain the average (second-order) behavior of the process.
- Example 2 (random phase) is WSS; Example 3 (random line) is not.

How SSS and WSS Relate

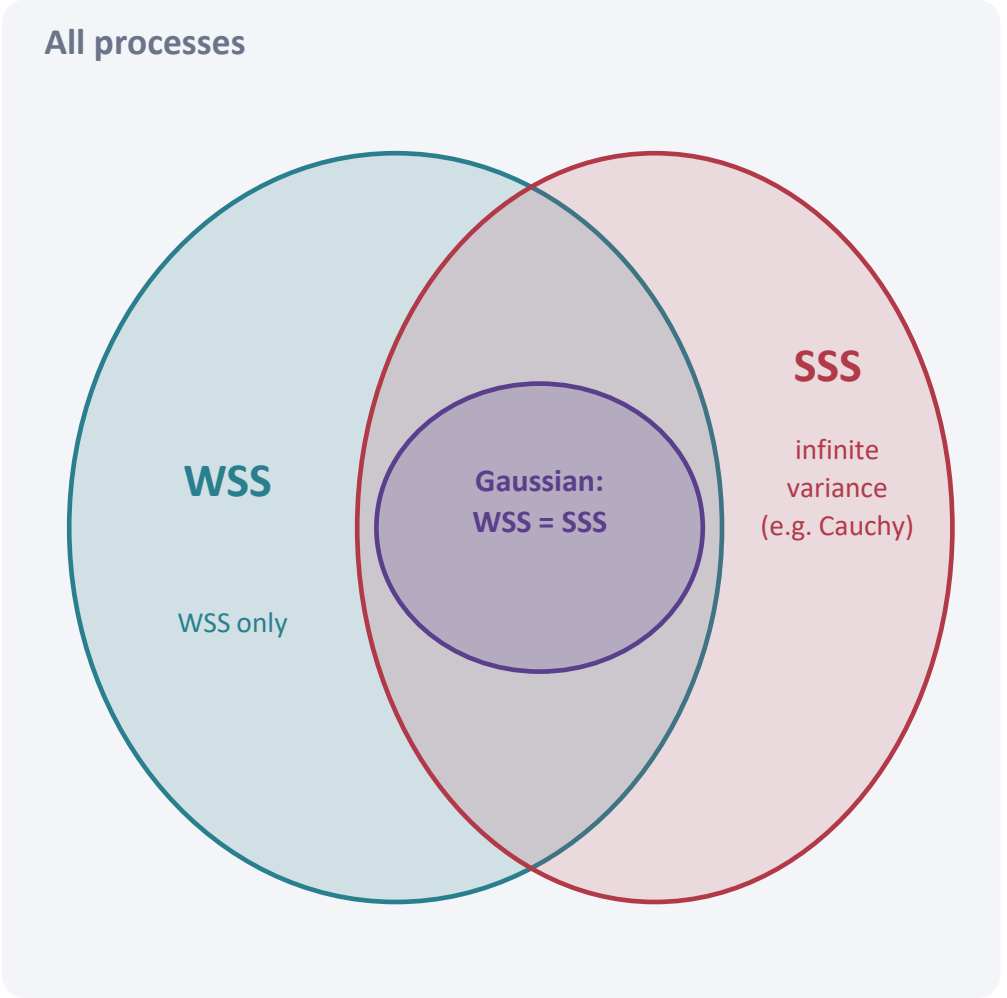
Facts

- **SSS ⇒ WSS**, provided the second moment is finite.
- **WSS ⇏ SSS** in general.
- For a **Gaussian process**: WSS ⇔ SSS.

Why the caveat?

An i.i.d. Cauchy sequence is SSS, but its mean and variance do not exist — so it is not WSS.

The Gaussian case is the key exception, because a Gaussian law is completely fixed by its mean and covariance (next slides).



A Process that is WSS but not SSS

$$X(t) = A\cos\omega_0 t + B\sin\omega_0 t$$

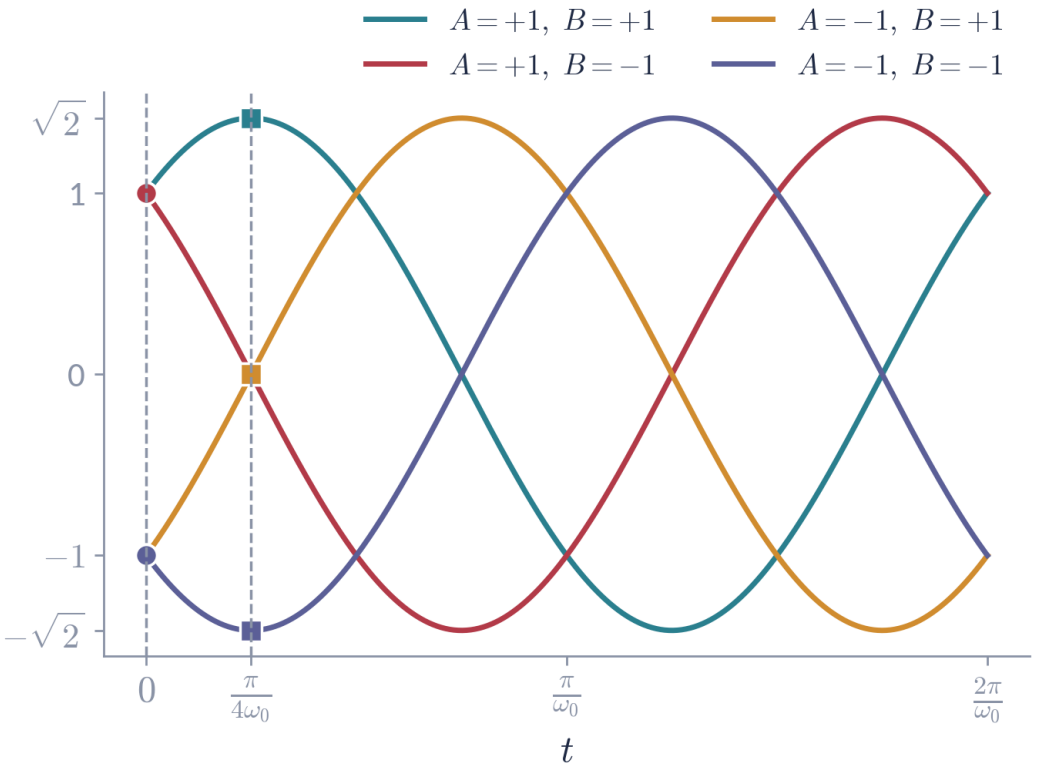
with A,B independent, each equal to ±1 with probability $1/2$ (zero mean, unit variance).

Using $E\{A^2\}=E\{B^2\}=1$ and $E\{AB\}=0$:

$$E\{X(t)\} = 0, \quad R_{XX} = \cos\omega_0 t_1 \cos\omega_0 t_2 + \sin\omega_0 t_1 \sin\omega_0 t_2$$

$$R_{XX}(t_1, t_2) = \cos\omega_0(t_1 - t_2) \;\Rightarrow\; \text{WSS}$$

- But at t=0: $X(0)=A\in\{-1,+1\}$.
- At $t=\pi/(4\omega_0)$: $X=(A+B)/\sqrt{2}\in\{-\sqrt{2}, 0, +\sqrt{2}\}$.
- The first-order pdf changes with $t \Rightarrow$ **not SSS** (not even first-order SSS).



The whole ensemble consists of four sample paths. Circles: values at t=0; squares: values at $t=\pi/(4\omega_0)$. With Gaussian A,B the same process would be SSS.

Characteristic Function (recall)

Characteristic function of a random variable

$$\Phi_X(\omega) = E\{e^{j\omega X}\} = \int_{-\infty}^{\infty} f_X(x)\, e^{j\omega x}\, dx$$

The Fourier transform of the pdf (sign convention $e^{+j\omega x}$); it determines the pdf uniquely.

Joint characteristic function of $X_1,\ldots,X_n$

$$\Phi_{\underline{X}}(\omega_1, \ldots, \omega_n) = E\left\{\exp(j \sum_{k=1}^{n} \omega_k X_k)\right\}$$

Determines the joint pdf of $X_1,\ldots,X_n$ uniquely.

Jointly Gaussian random variables

If $X_1,\ldots,X_n$ are jointly Gaussian with means μ_k and covariances C_{ik}=Cov(X_i,X_k):

$$\Phi_{\underline{X}}(\omega_1, \ldots, \omega_n) = \exp\left(j \sum_{k=1}^{n} \mu_k\, \omega_k - \frac{1}{2} \sum_{i=1}^{n} \sum_{k=1}^{n} C_{ik}\ \omega_i \omega_k \right)$$

A jointly Gaussian law is fixed completely by its means and covariances.

Gaussian Processes: WSS Implies SSS

If $X(t)$ is a Gaussian process, then by definition $X_1=X(t_1),\ldots,X_n=X(t_n)$ are jointly Gaussian for any n and any $t_1,\ldots,t_n$:

$$\Phi_{\underline{X}}(\omega_1,\ldots,\omega_n)=\exp\left(j\sum_{k=1}^{n}\mu_X(t_k)\,\omega_k-\frac{1}{2}\sum_{i=1}^{n}\sum_{k=1}^{n}C_{XX}(t_i,t_k)\,\omega_i\omega_k\right)$$

If $X(t)$ is also WSS, then $\mu_X(t_k)=\mu$ and $C_{XX}(t_i,t_k)=C_{XX}(t_i-t_k)$:

$$\Phi_{\underline{X}}(\omega_1,\ldots,\omega_n)=\exp\left(j\mu\sum_{k=1}^{n}\omega_k-\frac{1}{2}\sum_{i=1}^{n}\sum_{k=1}^{n}C_{XX}(t_i-t_k)\,\omega_i\omega_k\right)$$

Shifting all instants by c ($X_k'=X(t_k+c)$) leaves every t_i-t_k unchanged, so the joint characteristic function — and hence the joint pdf — of $\{X_k'\}$ is **identical** to that of $\{X_k\}$, for all n and all c.

Gaussian process: wide-sense stationary ⇔ strict-sense stationary. The Gaussian law depends only on second-order statistics, which is exactly what WSS fixes.

Example 1 Revisited — Integrator with WSS Input

Let $X(t)$ be a **zero-mean WSS** process and $z=\int_{-T}^{T} X(t)\,dt$. Then

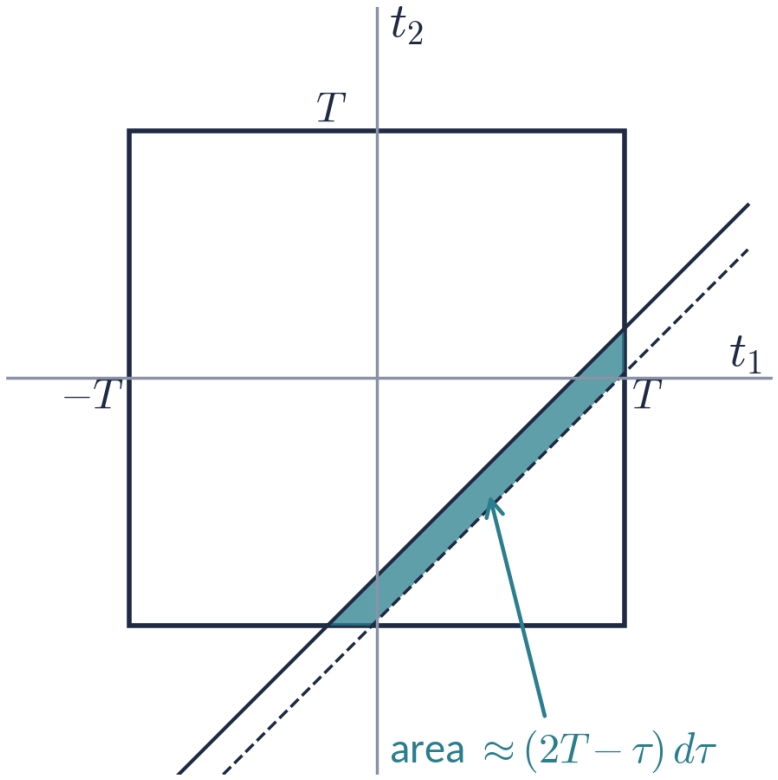
$$\sigma_z^2 = E\{|z|^2\} = \int_{-T}^{T}\int_{-T}^{T} R_{XX}(t_1 - t_2)\,dt_1\,dt_2$$

As t_1, t_2 range over $[-T, T]$, $\tau = t_1 - t_2$ ranges over $[-2T, 2T]$, and $R_{XX}(\tau)$ is constant on each strip $\tau < t_1 - t_2 < \tau + d\tau$. For $\tau > 0$ the strip has area

$$1/2\,(2T-\tau)^2 - 1/2\,(2T-\tau-d\tau)^2 \approx (2T-\tau)\,d\tau$$

(by symmetry, $(2T - |\tau|)\,d\tau$ in general). Hence the double integral reduces to a single one:

$$\sigma_z^2 = \int_{-2T}^{2T} R_{XX}(\tau)\,(2T-|\tau|)\,d\tau = 2T\int_{-2T}^{2T} R_{XX}(\tau)\left(1 - \frac{|\tau|}{2T}\right)d\tau$$



Autocorrelation of a Real WSS Process: Properties 1–2

1. Even symmetry

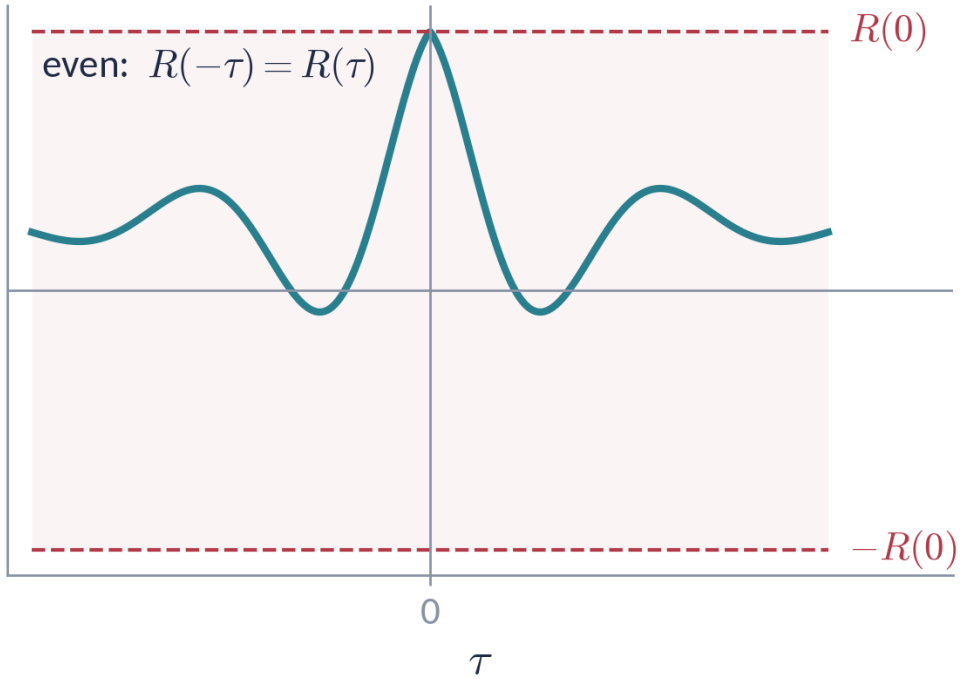
$$R_{XX}(-\tau) = E\{X(t-\tau)X(t)\} = R_{XX}(\tau)$$

(For complex processes: $R_{XX}(-\tau)=R_{XX}^{*}(\tau)$.)

2. Value at the origin = average power

$$R_{XX}(0) = E\{X^2(t)\} \geq 0$$

$$\sigma_X^2 = E\{X^2(t)\} - E\{X(t)\}^2 = R_{XX}(0) - \mu_X^2 = C_{XX}(0)$$



A typical autocorrelation: even, largest at τ=0, and confined to $[-R(0), R(0)]$ (property 3, next slide).

Autocorrelation of a Real WSS Process: Property 3

3. The maximum is at the origin

$$|R_{XX}(\tau)| \leq R_{XX}(0) \qquad \text{for all } \tau$$

Proof (Cauchy–Schwarz inequality for random variables, $|E\{UV\}|^2 \leq E\{U^2\}E\{V^2\}$), with $U=X(t+\tau)$, $V=X(t)$:

$$R_{XX}^2(\tau) = (E\{X(t+\tau)X(t)\})^2 \leq E\{X^2(t+\tau)\}\,E\{X^2(t)\} = R_{XX}^2(0)$$

Remarks

- The proof uses **ensemble averages** $E\{\cdot\}$, not time integrals: $R_{XX}(\tau)$ is defined through expectations.
- Consequently $|\rho_{XX}(\tau)| = |C_{XX}(\tau)|/C_{XX}(0) \leq 1$ for WSS processes.
- If $R_{XX}(\tau_0) = R_{XX}(0)$ for some $\tau_0 \neq 0$, then R_{XX} is periodic with period τ_0.

Cross-Correlation and Cross-Covariance

Two processes $X(t)$ and $Y(t)$

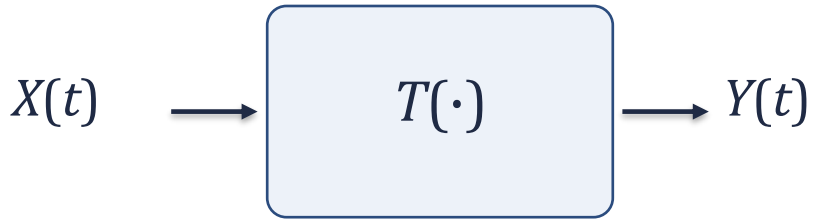
$$R_{XY}(t_1, t_2) = E\{X(t_1)\, Y^*(t_2)\}$$

$$C_{XY}(t_1, t_2) = R_{XY}(t_1, t_2) - \mu_X(t_1)\, \mu_Y^*(t_2)$$

Jointly WSS

$X(t)$ and $Y(t)$ are **jointly WSS** if each is WSS and $R_{XY}(t_1,t_2)$ depends only on $\tau=t_1-t_2$. Then

$$R_{XY}(\tau) = R_{YX}^*(-\tau), \qquad |R_{XY}(\tau)|^2 \le R_{XX}(0)\, R_{YY}(0)$$



Typical setting: $Y(t)$ is the output of a system driven by $X(t)$; R_{XY} tells us how input and output are related (Week 03: LTI systems).

Independent, Uncorrelated and Orthogonal Processes

Definitions

- **Independent:** $\{X(t_1),\ldots,X(t_n)\}$ and $\{Y(t_1'),\ldots,Y(t_m')\}$ are independent for all n,m and all time instants — their joint pdf factors.
- **Uncorrelated:** $C_{XY}(t_1,t_2)=0$, i.e. $R_{XY}(t_1,t_2)=\mu_X(t_1)\mu_Y^*(t_2)$, for all t_1,t_2.
- **Orthogonal:** $R_{XY}(t_1,t_2)=0$ for all t_1,t_2.

Independent ⇒ uncorrelated:

$$R_{XY}(t_1,t_2) = E\{X(t_1)\}\,E\{Y^*(t_2)\} = \mu_X(t_1)\,\mu_Y^*(t_2)$$

$$\Rightarrow\ C_{XY}(t_1,t_2) = \mu_X(t_1)\mu_Y^*(t_2) - \mu_X(t_1)\mu_Y^*(t_2) = 0$$

Careful

- The converse fails in general.
- It holds when X and Y are **jointly Gaussian**.
- Uncorrelated + one process zero-mean ⇒ orthogonal.

Example 4 — Second-Order Statistics of a WSS Process

$X(t)$ has mean and autocorrelation

$$\mu_X(t) = 3, \qquad R_{XX}(t_1, t_2) = 9 + 4e^{-0.2|t_1 - t_2|}$$

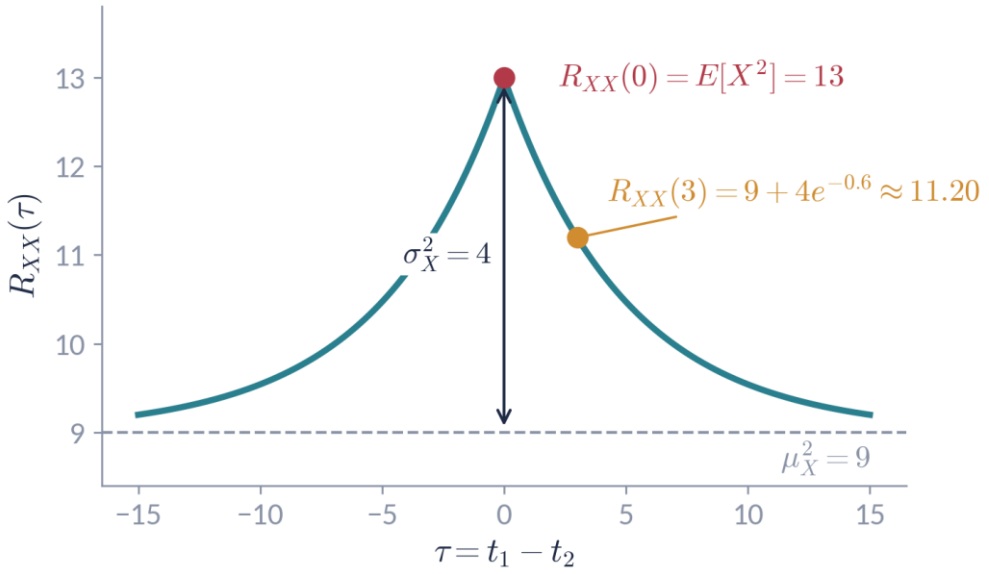
Constant mean and R_{XX} a function of $t_1 - t_2$ only ⇒ **WSS**. Let $Z = X(5)$ and $W = X(8)$.

$$E\{Z\} = E\{W\} = 3$$

$$E\{Z^2\} = R_{XX}(5,5) = R_{XX}(0) = 13$$

$$E\{W^2\} = R_{XX}(8,8) = R_{XX}(0) = 13$$

$$\mathrm{Var}(Z) = \mathrm{Var}(W) = 13 - 3^2 = 4$$



Example 4 — Continued

$$E\{ZW\} = E\{X(5)X(8)\} = R_{XX}(5,8) = 9 + 4e^{-0.2|5-8|} = 9 + 4e^{-0.6} \approx 11.195$$

$$\mathrm{Cov}(Z,W) = E\{ZW\} - E\{Z\}E\{W\} = 4e^{-0.6} \approx 2.195$$

$$\rho_{ZW} = \frac{\mathrm{Cov}(Z,W)}{\sigma_Z \sigma_W} = \frac{4e^{-0.6}}{4} = e^{-0.6} \approx 0.549$$

$$E\{(Z-W)^2\} = E\{Z^2\} + E\{W^2\} - 2E\{ZW\} = 26 - 2(9 + 4e^{-0.6}) \approx 3.61$$

Takeaway

Only the lag $|8-5|=3$ matters: $X(105)$ and $X(108)$ have exactly the same second-order statistics as $X(5)$ and $X(8)$.

Example 5 — The Poisson Process

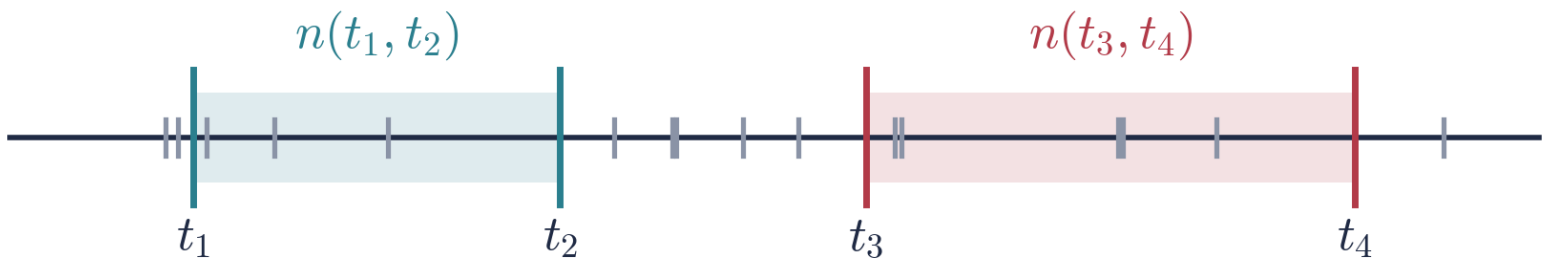
Consider a Poisson process with rate λ. Let $n(t_1,t_2)$ be the number of points in the interval $(t_1,t_2]$, $t_1 \leq t_2$.

Two defining properties

(a) The count in an interval of length t_2-t_1 is Poisson with parameter $\lambda(t_2-t_1)$:

$$P\{n(t_1,t_2)=k\} = e^{-\lambda(t_2-t_1)}\frac{[\lambda(t_2-t_1)]^k}{k!}, \qquad k=0,1,2,\ldots$$

(b) Counts in **non-overlapping intervals** are independent (independent increments).



non-overlapping intervals $\Rightarrow$ independent counts

Example 5 — Mean of the Counting Process

Define the counting process

$$X(t) = n(0,t), \qquad t \geq 0$$

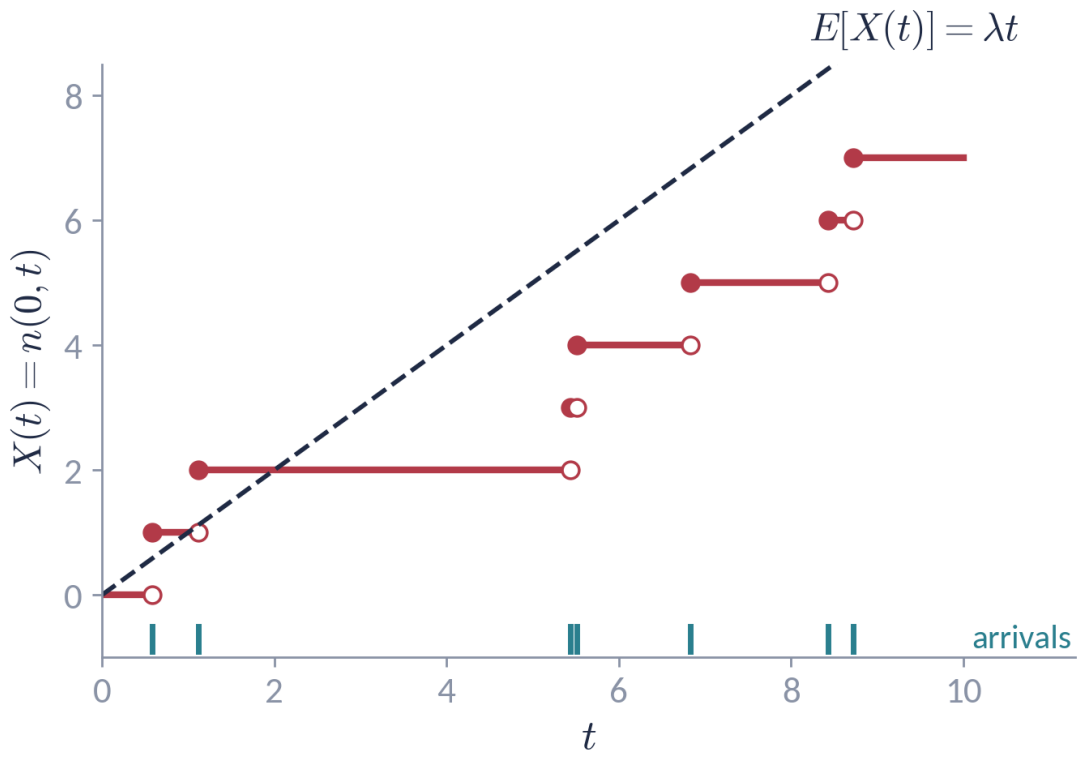
Since $n(0,t) \sim \text{Poisson}(\lambda\, t)$:

$$E\{X(t)\} = \lambda t$$

$$E\{X^2(t)\} = \text{Var} + \text{mean}^2 = \lambda t + \lambda^2 t^2$$

$$\sigma_X^2(t) = \lambda t$$

The mean grows linearly with t, so $X(t)$ is **not stationary** — not even first-order.



Example 5 — Autocorrelation of the Poisson Process

Suppose $t_1<t_2$ and split $n(0,t_2)=n(0,t_1)+n(t_1,t_2)$:

$$R_{XX}(t_1,t_2) = E\{n(0,t_1)\,n(0,t_2)\}$$
$$= E\{n^2(0,t_1)\} + E\{n(0,t_1)\,n(t_1,t_2)\}$$
$$= \lambda t_1 + \lambda^2 t_1^2 + E\{n(0,t_1)\}\,E\{n(t_1,t_2)\}$$
$$= \lambda t_1 + \lambda^2 t_1^2 + \lambda t_1 \cdot \lambda(t_2 - t_1) = \lambda t_1(1+\lambda t_2)$$

The third line uses **independent increments**: $n(0,t_1)$ and $n(t_1,t_2)$ count non-overlapping intervals.

In general (any order of t_1,t_2)

$$R_{XX}(t_1,t_2) = \lambda \min(t_1,t_2) + \lambda^2 t_1 t_2$$
$$C_{XX}(t_1,t_2) = \lambda \min(t_1,t_2)$$

Observe

R_{XX} is not a function of t_1-t_2 alone, so $X(t)$ is not WSS — even though its **increments** are stationary.

Example 6 — Waiting Time for a Taxi

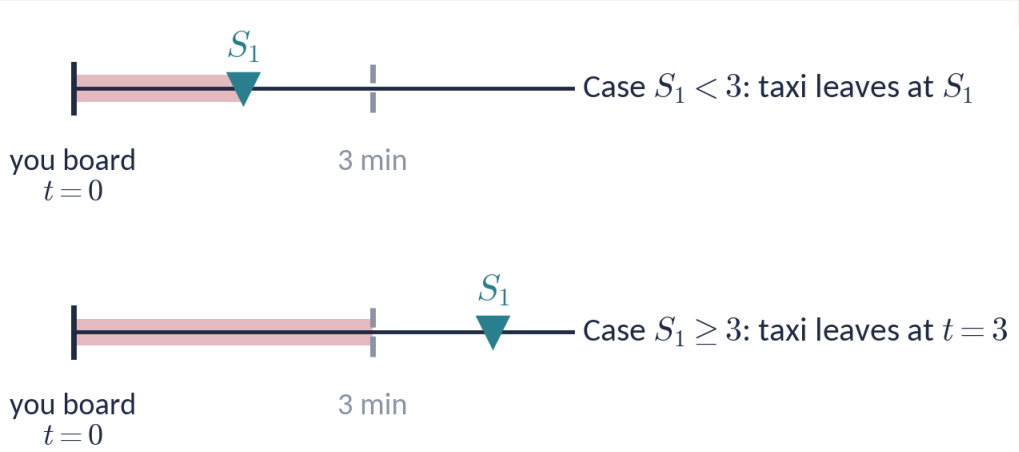
Problem

Taxis wait in a queue for passengers. Passengers arrive according to a Poisson process with an average of **60 passengers per hour**.

A taxi departs as soon as **two** passengers have been collected, or **3 minutes** after the first passenger got in — whichever comes first.

You get in a taxi as the first passenger. What is your **average waiting time** until departure?

Hint: condition on the first arrival after you get in the taxi.



Example 6 — Solution

With minutes as the unit of time, arrivals are Poisson with $\lambda=1$. Let S_1 be the arrival time of the next passenger: $S_1 \sim \text{Exp}(1)$, $P\{S_1>s\}=e^{-s}$. Your waiting time is $X=\min(S_1,3)$.

$$E\{X\} = E\{X \mid S_1 \geq 3\}P\{S_1 \geq 3\} + E\{X \mid S_1 < 3\}P\{S_1 < 3\}$$

$$= 3\,P\{n(0,3) = 0\} + \int_0^3 s\; f_{S_1}(s)\, ds = 3e^{-3} + \int_0^3 s\; e^{-s}\, ds$$

$$= 3e^{-3} + [-se^{-s}]_0^3 + \int_0^3 e^{-s}\; ds = 3e^{-3} - 3e^{-3} + 1 - e^{-3}$$

$$E\{X\} = 1 - e^{-3} \approx 0.950 \text{ min} \approx 57 \text{ s}$$

Shortcut

For a non-negative X, $E\{X\}=\int_0^\infty P\{X>s\}\, ds$:

$$E\{X\} = \int_0^3 e^{-s}\; ds = 1 - e^{-3}$$

Why Exp(1)?

By independent increments (memorylessness), the time from your boarding to the next arrival is exponential, whatever happened before.

Summary

	Strict-sense stationary	Wide-sense stationary
Condition	All n-th order pdfs invariant to time shift	μ constant and $R_{XX}(t_1,t_2)=R_{XX}(t_1-t_2)$
Involves	The complete probability law	First and second moments only
Checking it	Usually hard	Usually easy
Relation	SSS $\Rightarrow$ WSS (if $E\{\|X\|^2\}<\infty$)	WSS $\not\Rightarrow$ SSS, except for Gaussian processes
Examples	i.i.d. sequences; random-phase sinusoid	$A\cos\omega_0 t+B\sin\omega_0 t$ with uncorrelated A,B

NEXT WEEK

Ergodic Stochastic Processes

Stochastic Analysis of LTI Systems

When can time averages replace ensemble averages? What happens to $R_{XX}(\tau)$ when a WSS process passes through a filter?

Have a good day!